Introductory Mathematical Economics

Sequence

Delhi School of Economics

Sequence

Bound

Reading: Uploaded on Piazza

Sequence

Definition

A **sequence** is a function $f : \mathcal{N} \to \mathcal{R}$
That is f has domain $\mathcal{N}$ and range $\mathcal{R}$

$f(n)$ is the n-th term of the sequence
We shall denote a sequence by $\{a_n\}_1^\infty$. So $f(n) = a_n$.

Different ways of writing a sequence

(i) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$. Here $a_1 = 1, a_2 = \frac{1}{2}$ and so on

(ii) $\left\{\frac{1}{2^{n-1}}\right\}_{n=1}^{\infty}$. This means $a_n = \frac{1}{2^{n-1}}$

(iii) $a_1 = 1$, $a_{n+1} = \frac{1}{2}a_n$

Key Question

What do we want to know about a sequence?

Does the sequence 'end up somewhere' in the 'long run'

Examples of Sequence

- $a_n = \frac{1}{n}$
- $a_n = (-1)^n$
- $a_n = \sin(n)$
- $a_n = \frac{3n+8(-1)^n}{2n+5}$

Neighbourhood

Definition: Neighbourhood

For any $a \in \mathcal{R}$ and $\epsilon > 0$, the ϵ-**neighbourhood** of a is the set of all points whose distance from a is strictly less than ϵ. Formally,

$$B_\epsilon(a) = \{x \mid |x - a| < \epsilon\}$$

Convergence

Convergence of a sequence

A sequence $\{a_n\}_{n=1}^{\infty}$ **converges** to a, if it passes the following test:

Step 1: Choose a value of $\epsilon > 0$.

Step 2: There is a 'cut-off' N such that for all $n \geq N$, $a_n \in B_\epsilon(a)$

The sequence has to pass this test for all values of ϵ

Definition: A Closer Look

- 'for all $n > N$': captures the idea of 'long run'
- 'the sequence has to pass the test for all $\epsilon > 0$: captures the idea of 'getting close to the final destination

Observations

- Observation 1: In the test, you are free to choose different 'cut-off' N for different ϵ
- Observation 2: If a 'cut-off' N_1 works for ϵ_1 then it would work for any other $\epsilon_2 > \epsilon_1$
- Observation 3: The sequence may or may not actually touch the destination. That is there may not be any n such that $a_n = a$
- Observation 4: Removal of a finite number of points from the sequence does not affect its convergence

Formal Definition

A sequence $\{a_n\}_{n=1}^{\infty}$ **converges** to a if for every $\epsilon > 0$, there exists $N \in \mathcal{N}$ such that $\forall n \geq N$, $a_n \in B_\epsilon(a)$

Divergence

A sequence which does not converge to any $a \in \mathcal{R}$ is said to **diverge**

Special kind of divergence

A sequence $\{a_n\}_{n=1}^{\infty}$ goes to ∞ if for every $z > 0$, we can find N such that for all $n \geq N$, $a_n > z$.

Similarly, sequence $\{a_n\}_{n=1}^{\infty}$ goes to $-\infty$ if for every $z < 0$, we can find N such that for all $n \geq N$, $a_n < z$.

Results I

- Take a converging sequence $\{a_n\}_{n=1}^{\infty}$. If $a_n \leq z$ for all n, then $\lim_{n\to\infty} a_n \leq z$
- If weak inequality is replaced by strong inequality the inequality remains weak in the limit. That is $a_n < z$ for all n then $\lim_{n\to\infty} a_n \leq z$. It is possible that $a_n < z$ for all n but $\lim_{n\to\infty} a_n = z$.
- If $a_n \leq b_n$ for all n and both sequences converge then $\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n$
- If $a_n < b_n$ for all n, then $\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n$. It is possible that If $a_n < b_n$ for all n, but $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$

Results II

Let $\lim_{\to\infty} a_n = a$ and $\lim_{\to\infty} b_n = b$

- Take the sequence $\{ca_n\}_{n=1}^{\infty}$, where $c \in \mathcal{R}$. Then $\lim_{n\to\infty}(ca_n) = ca$.

- Take the sequence $\{a_n + b_n\}_{n=1}^{\infty}$. Then $\lim_{n\to\infty}(a_n + b_n) = a + b$.

Results III

Let $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$

- Take the sequence $\{c_n\}_{n=1}^{\infty}$ where $c_n = a_n b_n$. Then $\lim_{n\to\infty} c_n = ab$.
- Take the sequence $\{c_n\}_{n=1}^{\infty}$ where $c_n = \frac{a_n}{b_n}$. Moreover $b \neq 0$. Then $\lim_{n\to\infty} c_n = \frac{a}{b}$.

Bounded Sequence

Definition: Bounded Sequence

A sequence $\{a_n\}_{n=1}^{\infty}$ is **bounded** by $M \in \mathcal{R}$, if for all n, $|a_n| \leq M$.

Theorem: Every convergent sequence is bounded. However, the converse (every bounded sequence is convergent) is not true.

Monotone Sequence

Definition: Monotone Sequence

A sequence $\{a_n\}_{n=1}^{\infty}$ is **monotone** if it is increasing or decreasing. A sequence is increasing if $a_{n+1} \geq a_n$ for all n. Similarly a sequence is decreasing if $a_{n+1} \leq a_n \, \forall n$.

Theorem: If a sequence is increasing and bounded above then it converges (to its supremum)

To Summarize:

$$\text{Convergence} \Rightarrow \text{Bounded (but not monotonic)}$$
$$\text{Convergence to supremum} \Leftarrow \text{Bounded above and Increasing Convergence}$$
$$\text{to infimum} \Leftarrow \text{Bounded below and Decreasing}$$

Subsequence

Definition: Subequence

A subsequence $\{b_n\}_{n=1}^{\infty}$ of a sequence $\{a_n\}_{n=1}^{\infty}$ is a selection from the original sequence. That is $b_1 = a_{n_1}, b_2 = a_{n_2}, b_3 = a_{n_3}, \ldots$, where $n_1 < n_2 < n_3 < \ldots$

(ii) The subsequence itself must be a sequence

(i) Order cannot change

Subsequence

Theorem: All subsequence of a convergent sequence converges to the same limit.

Theorem: In a sequence, if we can find two subsequences that converge to different limits, then the original sequence diverges.

Theorem:If a sequence $\{a_n\}_{n=1}^{\infty}$ has two subsequences $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ such that

(i) $A = \{a_1, a_2, \ldots\} = X \cup Y$, where $X = \{x_1, x_2, \ldots\}$ and $Y = \{y_1, y_2, \ldots\}$ (we shall say these subsequences partition the original sequence)

(ii) $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ have the same limit (say z)

Then $\{a_n\}_{n=1}^{\infty}$ converges to z

Theorem: If all subsequences of a sequence $\{a_n\}_{n=1}^{\infty}$ converge to the same limit (say z), then $\{a_n\}_{n=1}^{\infty}$ also converges to z.

Sequence of Vectors

In a sequence of vectors $\{x_n\}_{n=1}^{\infty}$, each entry is a m-dimensional vector that is $x_n \in \mathcal{R}^m$.

$$x_1 = \begin{pmatrix} x_1^1 \\ x_1^2 \\ \vdots \\ x_1^m \end{pmatrix}, x_2 = \begin{pmatrix} x_2^1 \\ x_2^2 \\ \vdots \\ x_2^m \end{pmatrix}, \ldots, x_n = \begin{pmatrix} x_n^1 \\ x_n^2 \\ \vdots \\ x_n^m \end{pmatrix}, \ldots$$

Convergence of Sequence of Vectors

A sequence of vectors $\{x_n\}_{n=1}^{\infty}$ in $\mathcal{R}^m$ converges to $x \in \mathcal{R}^m$, if for every $\epsilon > 0$, there exists $N \in \mathcal{N}$ such that $\forall n \geq N,\ x_n \in B_\epsilon(x)$

where

$$B_\epsilon(x) = \left\{ y \middle| ||y - x|| < \epsilon \right\}$$

Convergence of Sequence of Vectors

Theorem: A sequence of vectors $\{x_n\}_{n=1}^{\infty}$ converges to $x \Leftrightarrow$ for all $k = 1, \ldots, m$, $\{x_n^k\}_{n=1}^{\infty}$ converges to x^k.

Proof of some Results

Let $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$

Result Take the sequence $\{ca_n\}_{n=1}^{\infty}$, where $c \in \mathcal{R}$. Then $\lim_{n\to\infty}(ca_n) = ca$.

Proof: If $c = 0$ then it is trivially true (check). Take $c \neq 0$. We want to show that for $\epsilon > 0$, we can find N such that for all $n \geq N$, $ca_n \in B_\epsilon(ca)$.
Lets choose that N for which $a_n \in B_{\frac{\epsilon}{|c|}}(a)$ for all $n \geq N$. Lets check that this N will indeed work. $|ca_n - ca| = |c||a_n - a| < |c|.\frac{\epsilon}{|c|} = \epsilon$.

Result Take the sequence $\{a_n + b_n\}_{n=1}^{\infty}$. Then $\lim_{n\to\infty}(a_n + b_n) = a + b$.

Proof: We want to show that for $\epsilon > 0$, we can find N such that for all $n \geq N$, $a_n + b_n \in B_\epsilon(a + b)$. The rest of the proof is about the construction of such N.
Pick N_1 such that $a_n \in B_{\frac{\epsilon}{2}}(a)$ for all $n \geq N_1$ (since $\lim_{n\to\infty} a_n = a$).
Pick N_2 such that $b_n \in B_{\frac{\epsilon}{2}}(b)$ for all $n \geq N_2$ (since $\lim_{n\to\infty} b_n = b$).
Take $N = \max\{N_1, N_2\}$. Lets check that this N will work
$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$

Result Take the sequence $\{a_n - b_n\}_{n=1}^{\infty}$. Then $\lim_{n\to\infty}(a_n - b_n) = a - b$.

Proof: Since $\lim_{n\to\infty}(ca_n) = ca$, we get $\lim(-a_n) = -a$ by choosing $c = -1$.
Now by using the sum of limit result, we obtain $\lim_{n\to\infty}(a_n - b_n) = a - b$.

Proof of some Results

Result Subsequence of a convergent sequence converges to the same limit.

Proof: $\{b_n\}_{n=1}^{\infty}$ is a subsequence of $\{a_n\}_{n=1}^{\infty}$ and $\lim_{n\to\infty} a_n = a$. We show that $\lim_{n\to\infty} b_n = a$. For every $\epsilon > 0$, we want to find N such that $\forall n \geq N$, $b_n \in B_\epsilon(a)$. Choose the N such that $n \geq N$, $a_n \in B_\epsilon(a)$. This N will do the job for b_n as well.

Result: If a sequence $\{a_n\}_{n=1}^{\infty}$ has two subsequences $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ such that

(i) $A = \{a_1, a_2, \ldots\} = X \cup Y$, where $X = \{x_1, x_2, \ldots\}$ and $Y = \{y_1, y_2, \ldots\}$ (we shall say these subsequences partition the original sequence)
(ii) $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ have the same limit (say z)

Then $\{a_n\}_{n=1}^{\infty}$ converges to z

Proof: Since $\{x_n\}_{n=1}^{\infty}$ converges to z, we can find N_1 such that $\forall n \geq N_1$, $x_n \in B_\epsilon(z)$. Similarly, since $\{y_n\}_{n=1}^{\infty}$ converges to z, we can find N_2 such that $\forall n \geq N_2$, $y_n \in B_\epsilon(z)$.
Choose $N = \max\{N_1, N_2\}$. For all $n \geq N$ both $x_n \in B_\epsilon(z)$ and $y_n \in B_\epsilon(z)$. Since all a_n is either in subsequence $\{x_n\}_{n=1}^{\infty}$ or in $\{y_n\}_{n=1}^{\infty}$, we have $a_n \in B_\epsilon(z)$ for all $n \geq N$. Hence proved.

Result: If all subsequences of a sequence $\{a_n\}_{n=1}^{\infty}$ converge to the same limit (say z), then $\{a_n\}_{n=1}^{\infty}$ also converges to z.

Proof: This follows immediately from the previous result. Since all subsequences converge to z, the same is true for subsequences that partition $\{a_n\}_{n=1}^{\infty}$.

Proof of some Results

Result Every convergent sequence is bounded. However the converse (every bounded sequence is convergent) is not true.

Proof: Take any sequence $\{a_n\}_{n=1}^{\infty}$ which converges to a. We can find N such that for all $n \geq N$, $|a_n - a| < 1$. Hence $|a_n| < |a| + 1$ (Triangle inequality).
Now choose $M = \max\{|a_1|, |a_2|, \ldots, |a_{N-1}|, (|a| + 1)\}$. Thus for all n, $|a_n| \leq M$. Proved.

Here is a bounded sequence which does not converge $1, -1, 1, -1, \ldots$

Result If a sequence is increasing and bounded then it converges (to its supremum).
Proof: Suppose $\{a_n\}_{n=1}^{\infty}$ is increasing and bounded.
Define $S = \{a_1, a_2, \ldots\}$. S is non-empty and bounded and hence has a supremum (completeness axiom). Let $s = \sup S$. We shall show that $\lim_{n\to\infty} a_n = s$. That is for every $\epsilon > 0$, we can find N such that for all $n \geq N$, $a_n \in B_\epsilon(s)$.
Take any $\epsilon > 0$. Since $s = \sup S$, there exist a_m such that $s > a_m > s - \epsilon$. As $\{a_n\}_{n=1}^{\infty}$ is an increasing sequence, for all $n > m$, $s > a_n \geq a_m > s - \epsilon$. Hence proved.

Proof of some Results

Result If a sequence $\{a_n\}_{n=1}^{\infty}$ converges to $a > 0$ then for any z between a and 0, there exists a N such that $a_n > z$ for all $n \geq N$

Proof Choose $\epsilon = a - z$, which is positive. Since $\{a_n\}_{n=1}^{\infty}$ converges to a, there is a N such that $a_n \in B_\epsilon(a)$ for all $n \geq N$. This implies $a_n > a - \epsilon = z$.

Result If a sequence $\{a_n\}_{n=1}^{\infty}$ converges to $a \neq 0$ then the sequence $\left\{\frac{1}{a_n}\right\}_{n=1}^{\infty}$ converges to $\frac{1}{a}$.

Proof Suppose $\{a_n\}_{n=1}^{\infty}$ converges to $a > 0$ (do the other case $a < 0$ yourself).
We have to show that for every $\epsilon > 0$ there exists a N such that

$$\left|\frac{1}{a_n} - \frac{1}{a}\right| < \epsilon \text{ for all } n \geq N$$

Note that, $\left|\frac{1}{a_n} - \frac{1}{a}\right| = \frac{|a_n - a|}{|a_n||a|}$.
Fix a $z \in (0, a)$. We can find N_1 such that $a_n > z$ for all $n \geq N_1$ (by previous result). Thus for $n \geq N_1$, $|a_n||a| > za$.
We can also find N_2 such that $|a_n - a| < za\epsilon$ because $\{a_n\}_{n=1}^{\infty}$ converges to a.
Therefore for all $n \geq \max\{N_1, N_2\}$, $\left|\frac{1}{a_n} - \frac{1}{a}\right| = \frac{|a_n - a|}{|a_n||a|} < \epsilon$. Hence proved.

Convergence of Sequence of Vectors

Result: A sequence of vectors $\{x_n\}_{n=1}^{\infty}$ converges to $x \Leftrightarrow$ for all $k = 1, \ldots, m$, $\{x_n^k\}_{n=1}^{\infty}$ converges to x^k. That is sequence of vectors converges if and only if component-wise sequences converge.

Proof Suppose $\{x_n\}_{n=1}^{\infty}$ converges to x. We want to show that component-wise sequences converge. That is $\{x_n^k\}_{n=1}^{\infty}$ converges to x^k.
Fix a $\epsilon > 0$. Since $\{x_n\}_{n=1}^{\infty}$ converges to x, we can find a N such that for all $n \geq N$ $||x_n - x|| < \epsilon$. That is, $\sqrt{\sum_{l=1}^{m}(x_n^l - x^l)^2} < \epsilon \Rightarrow \sum_{l=1}^{m}(x_n^l - x^l)^2 < \epsilon^2$. Since the sum of squares is less than ϵ^2 then each element of the addition is also less than ϵ^2. Thus $(x_n^k - x^k)^2 < \epsilon^2$ or $|x_n^k - x^k| < \epsilon$. Hence for all $n \geq N$, $|x_n^k - x^k| < \epsilon$. Thus $\{x_n^k\}_{n=1}^{\infty}$ converges to x^k.

Now suppose that $\{x_n^k\}_{n=1}^{\infty}$ converges to x^k for all $k = 1, \ldots, m$. Choose $\frac{\epsilon}{\sqrt{m}}$. Since $\{x_n^k\}_{n=1}^{\infty}$ converges to x^k, we get N^k such that for all $n \geq N^k$, we have $|x_n^k - x^k| < \frac{\epsilon}{\sqrt{m}}$ or $(x_n^k - x^k)^2 < \frac{\epsilon^2}{m}$.
For each dimension, we have identified a cutoff N_k that works for $\frac{\epsilon}{\sqrt{m}}$. Take $N = \max\{N_1, N_2, \ldots, N_k\}$. Thus for all $n \geq N$, $(x_n^k - x^k)^2 < \frac{\epsilon^2}{m}$ holds for all dimensions. Summing over dimension (that is k) we get, $\sum_{k=1}^{m}(x_n^k - x^k)^2 < \epsilon^2$, that is $||x_n - x|| < \epsilon$. Hence proved.